HW 0303

chapter 2

Question 1.

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a

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i=5$	$\sum y_i=10$	$\sum(x_i-\bar{x})=0$	$\sum(x_i-\bar{x})^2=10$	$\sum(y_i-\bar{y})=0$	$\sum(x-\bar{x})(y-\bar{y})=8$

平方

乘積

$\bar{x}=1$, $\bar{y}=2$

- 差距的大小
- 兩者之間的關係
- 一起變大or變小

for 斜率、相關性

b. OLS estimators

$$b_2=\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} \quad (2.7) \qquad b_1=\bar{y}-b_2\bar{x} \quad (2.8)$$

$$\hat{y}=1.2+0.8x$$

$$b_2=\frac{8}{10}=0.8 \qquad b_1=2-0.8\times 1=1.2$$

c

x	y	x^2	xy
3	4	9	12
2	2	4	4
1	3	1	3
-1	1	1	-1
0	0	0	0

$(x-\text{平均})^2$ 平方總合 - 平均的平方

- $\sum(x-\bar{x})^2=\sum x_i^2-N\bar{x}^2$

$\Rightarrow 10=15-5$

- $\sum(x_i-\bar{x})(y_i-\bar{y})=\sum x_iy_i-N\bar{x}\bar{y}$

$\Rightarrow 8=18-10$

$N=5$ $\sum_{i=1}^{5}x_i^2=15$ $\sum_{i=1}^{5}x_iy_i=18$

$N\bar{x}^2=5\cdot(1)^2=5$

$N\bar{x}\bar{y}=5\cdot1\cdot2=10$

d. $\hat{y} = 1.2 + 0.8x$ $\hat{e} = y - \hat{y}$

残差

x_i	Y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2.	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0

$\sum x_i = 5$ $\sum Y_i = 10$ $\sum \hat{y}_i = 10$ $\sum \hat{e}_i = 0^*$ $\sum \hat{e}_i^2 = 3.6$ $\sum x_i \hat{e}_i = 0^*$

残差平均0 残差与X不相关

1) 样本变异数

$$S_y^2 = \frac{\sum (y_i - \bar{y})^2}{N-1} = \frac{(4-2)^2 + (2-2)^2 + (3-2)^2 + (1-2)^2 + (0-2)^2}{5-1} = \frac{10}{4} = 2.5$$

$$S_x^2 = \frac{\sum (x_i - \bar{x})^2}{(N-1)} = \frac{10}{5-1} = 2.5$$

2) 样本共变异数

$$S_{xy} = \frac{\sum_{i=1}^{N} (y_i - \bar{y})(x_i - \bar{x})}{(N-1)} = \frac{8}{4} = 2$$

3) 样本相关係数

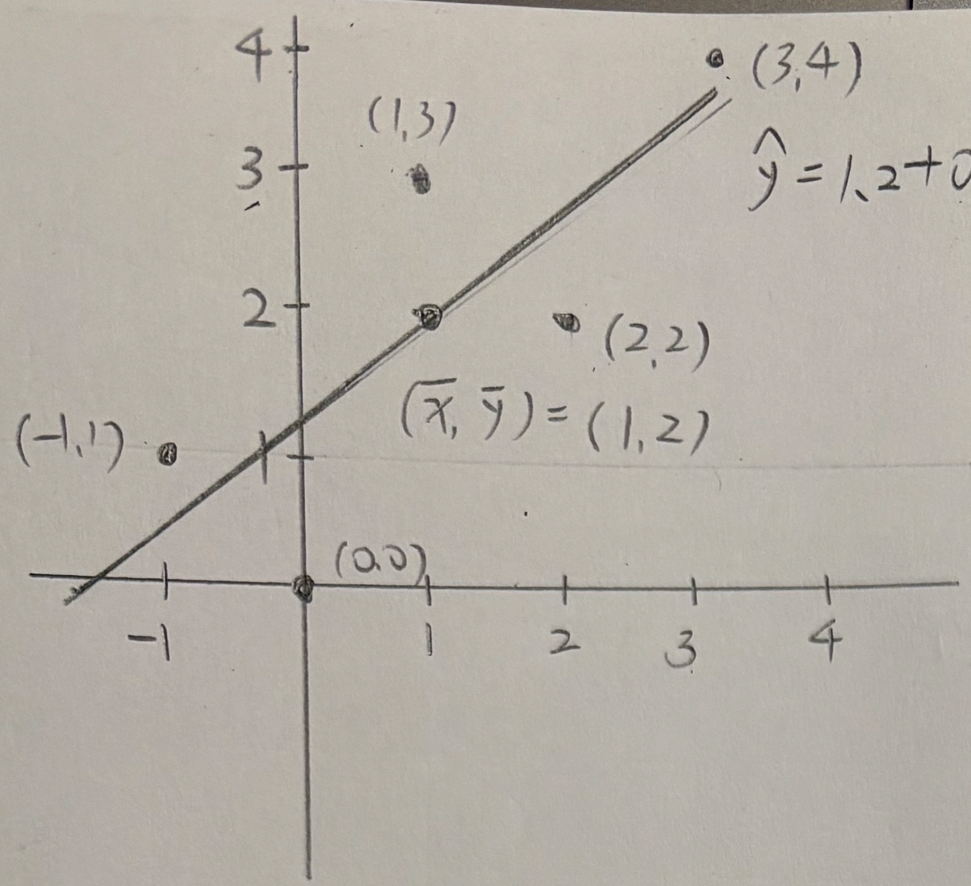
$$r_{xy} = \frac{S_{xy}}{(S_x S_y)} = \frac{2}{\sqrt{2.5} \times \sqrt{2.5}} = \frac{2}{2.5} = 0.8$$

4) 变异係数

$$CV_x = 100 (S_x / \bar{x}) = 100 \times \frac{\sqrt{2.5}}{1} \approx 100 \times 1.5811 \approx 158.11\%$$

X中位数：3, 2, 1, -1, 0 50th percentile, of $x = 1$

e



$\hat{y} = 1.2 + 0.8x$

OLS：求到線的距離平方總合最小

f $(\bar{x}, \bar{y}) = (1,2)$ 迴歸線通過 $(\bar{x}, \bar{y}) = (1,2)$

g $\bar{y} = b_1 + b_2\bar{x}$

$2 = 1.2 + 0.8 \times 1$

h $\overline{\hat{y}} = \bar{y}$ $\Sigma \hat{y}_i = 10$ $\overline{\hat{y}} = \frac{10}{5} = 2$ $\overline{\hat{y}} = \bar{y} = 2$

i $\hat{\sigma}^2$ = OLS 誤差變異估計

$$\hat{\sigma}^2 = \frac{\Sigma e_i^2}{N-2}$$ ← 估計2個參數 (b_1, b_2) 自由度 $N-K$

$$= \frac{3.6}{5-2}$$

$$= 1.2$$

j $\widehat{var}(b_2|X) =$ $Var(b_2|X) = \frac{\sigma^2}{\Sigma(x_i - \bar{x})^2}$

$$\widehat{var}(b_2) = \frac{\hat{\sigma}^2}{\Sigma(x_i - \bar{x})^2}$$

$$= \frac{1.2}{10} = 0.12$$

$Se(b_2) = \sqrt{0.12}$

標準誤 $= 0.346$